

RUVARASHE SHALOM MADZIME

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SUMMARY

Explainable AI researcher with peer-reviewed publications and hands-on software development experience. MSc thesis focuses on classifiers that produce logically grounded, human-readable explanations. Looking for positions where I can help teams understand, trust, and adopt AI products.

TECHNICAL SKILLS

Languages & Tools: Python, Java, SQL, Answer Set Programming (Clingo), SMT Solvers, Git, LaTeX

AI & Research: Explainable AI, Knowledge Representation, Defeasible Reasoning, NLP, LLMs, Machine Learning, SHAP

Other: Technical Writing, Data Visualization, Arduino, Asana, Microsoft Teams

EXPERIENCE

Research Assistant (SARCHI NRF), University of the Western Cape *Mar 2026 to Present*
Cape Town, South Africa

- Provide teaching support for undergraduate and postgraduate CS modules under Dr. Leenen's NRF SARCHI Chair grant.
- Co-supervise Honours students' research projects, giving feedback on methodology, writing, and experimental design.
- Conduct research in Knowledge Representation as part of the SARCHI programme.

Graduate Lecturing Assistant, University of the Western Cape *Feb 2025 to Nov 2025*
Cape Town, South Africa

- Ran tutorials and one-on-one consultations for CS modules, making complex topics accessible for 50+ students.
- Graded assignments and exams; supported lecturers with coursework design and student queries.

Graduate Research Assistant, University of the Western Cape *Jan 2024 to Nov 2024*
Cape Town, South Africa

- Built an LLM-based summarization system for Electronic Health Records, published at SACAIR 2024.
- Designed the full system from dataset analysis through to model evaluation using Python and ML tools.
- Helped secure research funding by reviewing existing literature and identifying gaps in EHR summarization.

Python Software Developer, Apparel Technology Inc. *Oct 2023 to Mar 2024*
Cape Town, South Africa

- Built a Python and Arduino monitoring system that tracked factory machine uptime and productivity in real time.
- Created a data visualization dashboard with Python Dash, giving operations teams live insight into machine performance.
- Managed project tasks and collaboration through Asana, Microsoft Teams, and Git.

EDUCATION

MSc Computer Science, University of the Western Cape & CAIR *Feb 2025 to Nov 2026*

- Thesis: Toward a Defeasible Semantics for Symbolic Classifiers.
- Area: Knowledge Representation, Defeasible Reasoning, Interpretable ML.

BSc (Hons) Computer Science, University of the Western Cape *Jan 2024 to Nov 2024*

- Cum Laude (85%). Modules: AI, Applied Data Science, IoT, Cybersecurity, Big Data Engineering.

BSc Computer Science, University of the Western Cape *Mar 2021 to Nov 2023*

- Cum Laude (84%). Key modules: Artificial Intelligence, Mathematics, Statistics.

PUBLICATIONS

- Madzime, R., & Meyer, T. (2026). *Toward Defeasible Machine Learning: When Learned Rule Sets Become Defeasible Theories*. In *Proceedings of NMR 2026: 24th International Workshop on Nonmonotonic Reasoning*.
- Madzime, R. S., Meyer, T., & Leenen, L. (2025). *An Override-Aware Classifier for Transparent AI*. *Proceedings of SACAIR 2025, Vol. II*.
- Madzime, R., & Nyirenda, C. (2024). *Enhanced Electronic Health Record Text Summarization Using Large Language Models*. *Proceedings of SACAIR 2024*.

HONORS & AWARDS

- UWC Mastercard Foundation Scholarship (2025 to 2026)
- Top Computer Science Student, University of the Western Cape (2023)
- KnowBe4 Women in Cybersecurity Scholarship (2023)
- Dean's Merit Award, University of the Western Cape (2021, 2022, 2023, 2024)